Lab 10: Epigenetics — Gene Regulation

BIOL-1

Name: ______________________________	**Date:** ______________________________

Overview

In this lab, you will explore how cells control which genes are turned on and off. You will model the lac operon (how bacteria regulate genes) and learn how epigenetic modifications control gene expression in human cells.

Materials

- This worksheet
- Colored pencils (optional)

Part 1: Why Gene Regulation Matters

1. Do all cells in your body have the same DNA? ____

2. Do all cells in your body make the same proteins? ____

3. Explain how a muscle cell and a brain cell can be so different even though they have the same DNA.

Part 2: The Lac Operon (Bacteria)

The lac operon is a group of genes in bacteria that make enzymes to digest lactose (milk sugar). The genes have an on/off switch.

When lactose is ABSENT:

- The repressor protein sits on the operator (the switch)
- RNA polymerase is blocked

- The genes are OFF — no enzymes are made

When lactose is PRESENT:

- Lactose binds to the repressor and changes its shape
- The repressor falls off the operator
- RNA polymerase can now read the genes
- The genes are ON — enzymes are made

4. What is the role of the repressor protein?

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5. What happens to the repressor when lactose is present?

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6. Why is this system energy-efficient for the bacterium?

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7. If a mutation makes the repressor unable to bind to the operator, will the genes be always on, always off, or normal? ____

8. If a mutation in the promoter prevents RNA polymerase from binding, will the genes be always on, always off, or normal? ____

Part 3: Epigenetics

Epigenetics = changes in gene expression that do NOT change the DNA sequence.

9. What is DNA methylation? Does it turn genes ON or OFF?

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10. What is histone acetylation? Does it turn genes ON or OFF?

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11. Fill in the table:

	DNA is...	Genes are...
Euchromatin	Loose / Tight?	On / Off?
Heterochromatin	Loose / Tight?	On / Off?

12. In a skin cell, the gene for keratin (skin protein) is active. Is the keratin gene in euchromatin or heterochromatin in this cell?

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13. In that same skin cell, the gene for hemoglobin (blood protein) is silent. Is the hemoglobin gene in euchromatin or heterochromatin?

Part 4: X-Inactivation

14. What is X-inactivation?

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15. What is a Barr body?

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16. Calico cats have patches of orange and black fur. The coat color gene is on the X chromosome. How does random X-inactivation create the patchy pattern?

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17. Why are calico cats almost always female?

Part 5: Big Picture

18. Name two environmental factors that can affect gene expression through epigenetics.

1.
2.

19. Identical twins have the same DNA. Why might they develop different diseases as they get older?

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20. What is the key difference between a genetic mutation and an epigenetic change?
